

Production Target Optimization — 6D Variant

Reduced search space; profile-only Pareto BO

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Mu2e — autoresearch / closed-loop BO

Status (2026-06-12): pt6d01 (10 Sobol) + pt6d02 R0 (10 qNEHVI) + R1 partial (7 so far) = **27 evals**. **Headline:** `pt6d02R00_06` at $\mu=2.320\times 10^{-3}$ (+7.1%) / dose= 1.300×10^{-9} Gy/POT (−14%) — a Stickman dominator in qNEHVI's first batch, matching the 10D campaign's best dominator after 3 rounds.

What we drop vs the 10D variant

dropped knob	10D range	6D treatment
N — plate count	25 – 45 (Integer)	fixed at 35 (Stickman count)
l0, l1, l2 — lug profile	4 – 12 mm	derived: $lug = t_{Plate} + 0.75\text{ mm}$

What remains:

knob (3 each)	range	units
r0, r1, r2 — r0ut profile	2.0 – 4.5	mm
t0, t1, t2 — plate thickness	3.0 – 7.0	mm

t-upper capped at 7.0 (not 8.0) to dodge the thick-plate spacer-overlap regime documented in `prodtarget-spacer-supportring-overlap`. Lug-on-spacer geometry derives from `t` directly, so no free lug dimension is needed.

Why a 6D track

The 10D campaign has produced 39 evals across pt001 → ptX05; best simultaneous Stickman-beat is `ptX05R02_07` at $\mu=2.31 \times 10^{-3}$ / dose= 1.39×10^{-9} Gy/POT. Open questions the 6D variant probes:

- **Does the lug profile matter for thermal performance?** If picks consistently come back with `l ≈ t + small offset`, the integer + 3 lug dims in the 10D variant are noise — 6D would converge faster per CPU-hour spent.
- **Does varying `N` add Pareto coverage, or just rediscover knee points?** 10D best sat at `N=35,36` (Stickman count anyway). Fixing N tests whether the integer-axis exploration was load-bearing.
- **Cleaner Pareto** — same `(μ, peak_dose)` objectives; reduced confounding from N+lug make the front easier to read on the slide deck.

Optimizer & wiring (identical to 10D)

- **qLogNoisyExpectedHypervolumeImprovement** — multi-obj Pareto BO, log-stabilized near saturation.
- `q=10` per round, sequential-greedy via batched fantasies.
- **Cold-start fix** (2026-06-10): `botorch_predict._sobol_cold_start` draws Sobol points when history < 2. pt6d01 launched cleanly with empty leaderboard, no priors-projection.
- **Mode wiring** = MODE_SPECS entry + 6-knob `ProdTarget6DMode.build_space` + lug-derivation in `_expand`; reuses the entire `pot_only` / `harvest` / `peak_dose_Gy_per_P0T` path.

`pt6d02` is warm-started on pt6d01's 10 Sobol rows; first true qNEHVI fit-and-pick happened at launch (R0).

GP cloud — pt6d01+pt6d02 landscape (n=27)

65 536 Sobol-sampled posterior means; GP trained on n=27 (pt6d01 cold-start + pt6d02 R0+R1). Black star = **Stickman baseline**; magenta stars = 6D evals.

- Observed μ range $1.75 - 2.32 \times 10^{-3}$; dose range $1.30 - 3.89 \times 10^{-9}$ — qNEHVI explored well above Stickman dose for hypervolume coverage.
- **Dominator:** **pt6d02R00_06** at $\mu=2.320 \times 10^{-3}$ / dose= 1.300×10^{-9} — beats Stickman on BOTH axes (+7.1% μ , -14% dose) in qNEHVI's first batch.
- GP posterior envelope now spans $1.81 - 2.29 \times 10^{-3}$ predicted μ , $1.3 - 4.0 \times 10^{-9}$ predicted dose — large expansion from the n=10 cold-start view; picker has a real surface to chase.

Status & next steps

Now (27 evals): pt6d01 R0 (10 Sobol) + pt6d02 R0 (10 qNEHVI) + R1 (7 so far; rest still grinding). `pt6d02R00_06` is the headline: Stickman-dominator on first qNEHVI batch.

Answer to the open question: the 6D variant matched the 10D campaign's best simultaneous Stickman-beat (`ptX05R02_07` at $\mu=2.31e-3$ / dose= $1.39e-9$) in the very first qNEHVI batch — vs 3 full rounds (30 evals) in 10D. Lug + N axes were noise for thermal-coupled optimization around Stickman's natural sweet spot.

Deferred (shared with 10D):

- **Path B NIEL** as 3rd objective — neutron channel still missing.
- `plateMaterial` categorical — needs mixed-model picker.
- Closed-loop final-round orphan-on-exit bug — same barrier-timeout-zero-rows-falsepos class as 10D campaign.